

# Muon $g - 2$ : Status 2025

Fermilab final result, White Paper 2025 and assessment of the FFGFT statements

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# Muon $g - 2$ : Status 2025

## Abstract

Several early documents of the corpus refer to the deviation between the measured anomalous magnetic moment of the muon and the Standard Model prediction as it stood in 2021:  $\Delta a_\mu = 251(59) \times 10^{-11}$ , about  $4.2 \sigma$ . With the final result of the Fermilab experiment (June 2025) and the 2025 White Paper of the Muon  $g - 2$  Theory Initiative, this deviation has shrunk to  $38(63) \times 10^{-11}$ , about  $0.6 \sigma$  **[K]**. Statements that cite a T0 contribution of  $251 \times 10^{-11}$  as the explanation of the anomaly are therefore outdated; such a fixed contribution would today lie about  $3.4 \sigma$  above the difference **[K]**.

The load-bearing  $g - 2$  line of the corpus (Doc. 018 Rev. 12, Doc. 158, A138) is not affected. It rests on *ratios*:  $\Delta a(\tau - \mu) / \Delta a(\mu - e) = f^{1/3} - 1$  and the bridge  $\Delta a(\tau - e) / \Delta a(\mu - e) = (144/125) m_\tau / m_\mu$ . In these, the projection factor and the SI conversion cancel. It obtains absolute values by anchoring the ratios to measured quantities; with the 2025 values this gives  $a_\tau = 1.2811 \times 10^{-3}$  **[K]**. This anchored value is the better absolute value compared with the direct formula, and it depends only on the measured values of  $a_e$  and  $a_\mu$ , not on the Standard Model calculation, whose status has changed.

## Status markers

- [K] Numerically confirmed (verification script).  
 [S] Speculative: a hint or conjecture, not yet derived or checked in the corpus.

The experimental and theoretical input values are taken from the sources listed in the bibliography; they are not derived here.

## 1 Starting point in the corpus

The corpus contains two kinds of statements on the lepton  $g - 2$ , which have to be kept apart.

**(a) Explaining the anomaly by a T0 contribution.** Early documents posit a contribution  $\Delta a_\mu^{(T0)} = 251 \times 10^{-11}$ , or describe the muon anomaly as explained by the  $\xi$  term, or use it as a benchmark: Doc. 019, 030, 033, 049, 053, 059, 070, 081, 095, and in the A series A138 (comparison with the "muon discrepancy"). These statements depend directly on the size of the deviation between measurement and Standard Model.

**(b) The ratio line.** Doc. 018 (Rev. 12), Doc. 158 and A138 describe the leptons by fractal winding exponents ( $5/3$  for the muon,  $4/3$  for the tau) and put the ratios at the centre: ratios of  $g - 2$  differences are structurally exact, because the projection factor, the fractal correction and the SI conversion cancel. Absolute values follow only by anchoring to a measured quantity – the same division of labour as for the masses, where ratios are fixed geometrically and one empirical anchor sets the scale (A138, Doc. 351). This line compares with measured values, not with the difference to the Standard Model.

## 2 Status 2025

- **Measurement.** The Fermilab experiment E989 published its final result in June 2025:  $a_\mu = 116\,592\,070.5(14.8) \times 10^{-11}$ , a precision of 127 ppb [1]. Together with E821 (Brookhaven) this gives the world average  $a_\mu^{\text{exp}} = 116\,592\,071.5(14.5) \times 10^{-11}$  [2].
- **Theory.** The 2025 White Paper of the Muon  $g - 2$  Theory Initiative gives  $a_\mu^{\text{SM}} = 116\,592\,033(62) \times 10^{-11}$  [2]. The essential change compared with 2020 is the hadronic vacuum polarisation: it is now determined from lattice QCD instead of  $e^+e^-$  cross sections.
- **Difference.**  $a_\mu^{\text{exp}} - a_\mu^{\text{SM}} = 38(63) \times 10^{-11}$ , about  $0.6\sigma$  [K]. In 2021 it was  $251(59) \times 10^{-11}$ , about  $4.2\sigma$  [K].
- **Open within the theory.** Why the data-driven determination of the hadronic vacuum polarisation differs from the lattice determination is not yet understood [3]. The anomaly is therefore regarded as strongly weakened but not definitively settled.

What has changed is thus the Standard Model calculation, not the measured value: the world average of  $a_\mu$  has moved by only about  $10 \times 10^{-11}$  since 2021.

## 3 Consequences for statements of type (a)

A fixed T0 contribution of  $251 \times 10^{-11}$  was, in 2021, exactly as large as the deviation it was meant to explain. Today it lies

$$\frac{251 - 38}{63} \approx 3.4\sigma$$

above the measured difference **[K]**. The statement “the  $\xi$  term explains the muon anomaly” is therefore outdated. For an additional contribution beyond the Standard Model, the current status leaves a range of about  $-88$  to  $+164 \times 10^{-11}$  at two standard deviations **[K]**. A T0 contribution of this size is not excluded, but it is not derived in the corpus **[S]**.

The documents listed under (a) in Section 1 are affected. They carry a note after the abstract that refers to this document; their text is otherwise unchanged.

## 4 The ratio line is not affected

The core of Doc. 018, Doc. 158 and A138 is two ratio statements **[K]**:

$$\frac{\Delta a(\tau-\mu)}{\Delta a(\mu-e)} = f^{1/3} - 1 = 18.57, \quad \frac{\Delta a(\tau-e)}{\Delta a(\mu-e)} = \frac{144}{125} \frac{m_\tau}{m_\mu} = 19.37.$$

The first follows from the exponents  $5/3$  and  $4/3$  alone. The second replaces  $f^{1/3} = 19.57$  by the ratio of the mass formulas and thereby links  $g - 2$  to the lepton masses;  $f$ ,  $k_{\text{geom}}$  and  $K_{\text{frak}}$  cancel.

**From ratio to absolute value.** Anchoring the bridge to the measured values of  $a_e$  and  $a_\mu$  gives the absolute value for the tau:

$$a_\tau = a_e + \frac{144}{125} \frac{m_\tau}{m_\mu} (a_\mu - a_e) = 1.2811 \times 10^{-3}$$

with the 2025 values **[K]**; Doc. 018 and A138 quote  $1.282 \times 10^{-3}$  with a rounded  $a_e$ , Doc. 158  $1.281 \times 10^{-3}$ . The direct absolute formula  $a_\tau = a_e + 4\pi/f^{4/3}$  gives  $1.2454 \times 10^{-3}$  instead **[K]**. The anchored value is the better one, because it contains only the exact ratio and measured quantities; this is why Doc. 158 and A138 use it as the prediction.

**Why the anomaly plays no role here.** The anchor is the measured difference  $a_\mu - a_e = 6.2685 \times 10^{-6}$ . It has changed by only about  $10^{-10}$  between 2021 and 2025 **[K]**. The anomaly was a statement about the Standard Model calculation; the ratio line does not use that calculation. The tau prediction therefore remains unchanged. It lies  $1.04 \times 10^{-4}$  above the Standard Model prediction  $a_\tau^{\text{SM}} = 1.1772 \times 10^{-3}$  **[K]**. The current experimental bounds on  $a_\tau$  are about 600 times wider than this difference **[K]**; the decision awaits a measurement of  $a_\tau$  **[S]**.

## 5 The direct absolute formulas

Besides the ratio line, Doc. 018 contains direct absolute formulas from the winding geometry, corrected by  $K_{\text{frak}}^{3/2}$ . With the values of Doc. 018 one obtains **[K]**:

$$a_e^{(\text{corr})} = 1.1598 \times 10^{-3} \quad (0.014 \% \text{ from the measured value}),$$

$$a_\mu^{(\text{corr})} = 1.1642 \times 10^{-3} \quad (-0.15 \% \text{ from the world average}).$$

These values also compare with measured values and do not depend on the anomaly. They work at the level of  $10^{-4}$  to  $10^{-3}$ , however, and are not a precision comparison where the anomaly was discussed. A numerical example shows why the corpus takes the route via ratios for absolute values: the directly computed difference  $\Delta a(\mu-e) = 4\pi/f^{5/3} = 4.373 \times 10^{-6}$  lies about 30 % below the measured one **[K]**, while the ratio of the differences follows exactly from the exponents. The anchored route avoids this uncertainty.

**A138.** The comparison there, that the tau deviation is “seven times larger than the muon discrepancy”, has become moot, since there is no longer a significant muon discrepancy. The tau statement itself is independent of this.

## 6 Result

With the 2025 status, the muon  $g - 2$  anomaly of 2021 is no longer significant. FFGFT statements that use it as evidence or as the target of a T0 contribution are outdated and are marked in the affected documents. The load-bearing ratio line is not affected: its ratios are structurally exact, it obtains its absolute values by anchoring to measured quantities, and the measured values have not changed. Its tau prediction  $a_\tau \approx 1.281 \times 10^{-3}$  remains the actual test statement. Whether the FFGFT additionally yields a contribution to the muon within today's allowed range of about  $10^{-9}$  is open **[S]**.

Verification script: `2/python/Dok382_Skripte/pruef_382_myon_g2_stand.py`, 21/21.

# Bibliography

- [1] Muon  $g - 2$  Collaboration (D. P. Aguillard et al.), *Measurement of the Positive Muon Anomalous Magnetic Moment to 127 ppb*, Phys. Rev. Lett. 135 (2025) 101802, arXiv:2506.03069.
- [2] R. Aliberti et al. (Muon  $g - 2$  Theory Initiative), *The anomalous magnetic moment of the muon in the Standard Model: an update*, Phys. Rep. 1143 (2025) 1–158, arXiv:2505.21476.
- [3] CERN Courier, *Fermilab's final word on muon  $g-2$*  (2025).